

# Multi-Rail Isolated Power Supply Monitoring System

## Capstone Design Documentation

Michael Capotosto

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# Chapter 1

## Introduction

### 1.1 Project Overview

The Multi-Rail Isolated Power Supply Monitoring System is a hardware and software system developed to provide centralized monitoring of multiple electrically isolated power-supply rails. The system measures both output voltage and load current, provides configurable alarm limits, displays operating status and measurement data via a local display or ethernet for integration with an EPICS-based control system.

The completed system monitors six independent power-supply rails:

- +5VA
- +5VB
- +5V
- -5V
- +15V
- -15V

Because the monitored supplies may be electrically isolated from one another, the monitoring system must preserve that isolation while still allowing all six channels to communicate with a common controller. Each supply rail therefore uses an independent voltage/current monitoring channel and an isolated digital communication path.

The primary controller is an Arduino GIGA R1 WiFi. The controller acquires measurement data from the individual monitor channels, converts the data to engineering units, applies the appropriate rail polarity, evaluates configured alarm limits, and presents the resulting information on a GIGA Display Shield. An Ethernet interface provides network communication between the embedded controller and an EPICS Input/Output Controller (IOC). The EPICS IOC then makes the measurements, alarms, and system status available to a CS-Studio Phoebus operator interface.

## 1.2 Project Motivation

Power-supply systems containing multiple isolated voltage rails present a monitoring challenge because each circuit associated with one supply can't share a direct electrical reference with another supply. Connecting conventional measurement or communication circuits across these references can break isolation between the rails.

A centralized monitor is desirable because it allows the user to observe all supply voltages and currents from a single interface, identify abnormal operating conditions, detect communication or sensor faults, and integrate the power-supply status into a larger control system.

The design developed for this project addresses these requirements by performing voltage and high-side current measurement locally on each monitored rail and transferring the resulting information across electrically isolated digital interfaces. The controller receives measurement data from all six rails without requiring their power references to be directly connected together.

## 1.3 System Architecture

The monitoring system is divided into three primary functional areas: the power-supply monitoring hardware, the microcontroller, and the supervisory control interface.

The power supply monitoring hardware contains the current-sense resistors, LTC2945 PMICs, and digital isolation circuitry. Each LTC2945 measures the supply voltage and the differential voltage developed across a high-side current-sense resistor. The resulting digital measurement information is sent over the I<sup>2</sup>C bus while maintaining the required electrical isolation.

The Arduino GIGA R1 WiFi provides the mcu processing functions. Its firmware initializes the monitoring channels, acquires voltage and current measurements, detects communication errors, converts raw measurements to engineering units, applies channel-specific polarity and scaling, evaluates alarm limits, and updates the local display. Configuration information, including network parameters and alarm limits, is retained in persistent storage so that required settings can survive a controller reset or power cycle.

Network communication is provided through an Ethernet interface using the PSC protocol. The EPICS IOC communicates with the embedded controller using PSCDriver and converts the transmitted values into EPICS process variables. CS-Studio Phoebus provides the remote operator interface for viewing measurements, alarm conditions, connection status, and system state.

## 1.4 Design Objectives

The major design objectives for the Multi-Rail Isolated Power Supply Monitoring System are to:

1. Measure the voltage of six independent power-supply rails.
2. Measure the load current of each rail using high-side current sensing.
3. Preserve electrical isolation between independently referenced power-supply rails and the common controller.
4. Convert acquired data into meaningful engineering units and correctly represent both positive and negative voltage rails.
5. Detect I<sup>2</sup>C communication failures and prevent invalid sensor data from being presented as valid measurements.
6. Compare voltage and current measurements against configurable upper and lower alarm limits.
7. Provide a local graphical display of voltage, current, alarm status, communication status, and network state.
8. Retain configuration and alarm settings through controller resets and power cycles.
9. Provide Ethernet communication for remote monitoring and configuration.
10. Integrate the monitoring system with an EPICS IOC and CS-Studio Phoebus operator interface.
11. Maintain the hardware, firmware, and control-system software under revision control so that changes made during design, implementation, and testing remain traceable.

## 1.5 Project Scope

The scope of the project includes the design and implementation of the sensor interface hardware, Arduino GIGA controller hardware, embedded firmware, Ethernet communication interface, EPICS IOC, and operator-interface software. The hardware portion includes the six voltage/current monitoring channels, current-sense resistors, digital isolation circuitry, controller interface, Ethernet circuitry, and supporting power-supply circuitry.

The software portion includes the embedded acquisition and alarm-processing firmware, persistent configuration storage, local display interface, PSC network communication, EPICS IOC configuration and database records, and the CS-Studio Phoebus operator interface.

Design files and software associated with the project are maintained in the project Git repository:

[https://github.com/capotosto/PS\\_Monitor](https://github.com/capotosto/PS_Monitor)

The repository provides revision control for the hardware design, firmware, EPICS IOC files, documentation, and other project artifacts generated during development.

## 1.6 Report Organization

This report documents the development of the monitoring system from preliminary design through implementation and testing. The preliminary-design chapter defines the major functional blocks, presents the system architecture, documents supporting engineering calculations, includes the hardware schematics and bills of materials, and describes the primary firmware and control-system software flows.

Subsequent chapters document prototyping and development activities, the EPICS IOC implementation, and the CS-Studio Phoebus operator interface. Supporting project-management and design documents developed during the earlier phases of the capstone project are retained as part of the overall project documentation and provide traceability between the original requirements, design decisions, implementation, and final system.

# Chapter 2

## Preliminary Design (Week 1 Assignment)

### 2.1 Purpose and Scope

This chapter documents the preliminary hardware and software design of the Multi-Rail Isolated Power Supply Monitoring System. The design monitors six power-supply rails, measures rail voltage and load current, preserves electrical isolation between floating supplies and the controller, evaluates measurements against user-configurable alarm limits, presents local status on the controller display, and publishes measurements and status to an EPICS control-system interface.

This preliminary design is intended to establish the major analog and digital functional blocks, document supporting engineering calculations, define the hardware schematic base-line, and provide software flowcharts detailed enough to support implementation. Simulation and additional verification results can be added in the subsequent design phase.

### 2.2 Major Functional Blocks

The major system functions are summarized below.

1. **Power-Supply Interface:** Connect to the six monitored rails and route each rail through the associated voltage/current monitoring channel.
2. **Current Measurement:** Convert load current to a differential voltage across a precision shunt resistor and digitize it with the LTC2945.
3. **Voltage Measurement:** Measure the monitored rail voltage using the LTC2945 supply-voltage ADC channel.
4. **Electrical Isolation:** Maintain galvanic isolation between each floating monitored supply and the common controller-side I<sup>2</sup>C bus.

5. **Digital Acquisition:** Poll each monitor, identify communication faults, apply channel-specific sign/scaling information, and calculate engineering-unit values.
6. **Alarm Processing:** Compare voltage and current values against configured upper and lower limits and generate channel alarm states.
7. **Local User Interface:** Present all rail measurements, alarms, network state, and communication status on the Arduino GIGA Display.
8. **Network/EPICS Interface:** Publish measurements, alarm limits, alarm state, and connection status over Ethernet for the EPICS IOC and Phoebebus interface.

## 2.2.1 Functional Block Diagram

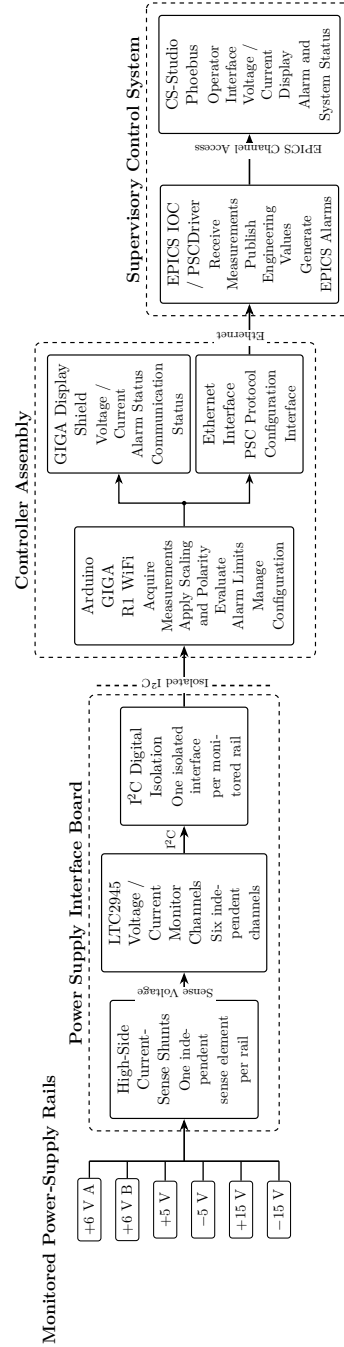


Figure 2.1: Preliminary functional block diagram of the multi-rail isolated power supply monitoring system. Six independent power-supply rails are measured using high-side current sensing and LTC2945 voltage/current monitoring channels. Digital isolation preserves electrical independence between the monitored supplies and the common controller. The Arduino GIGA processes the measurements, provides a local operator display, and transmits measurement and status information over Ethernet to the EPICS IOC and CS-Studio Phoebe operator interface.

## 2.3 Preliminary Engineering Calculations

### 2.3.1 Current-Sense Shunt Resistor Sizing

The LTC2945 measures the differential voltage between SENSE+ and SENSE– with a full-scale range of 102.4 mV. The ideal shunt resistance for a specified maximum measurable current is therefore

$$R_{SHUNT,ideal} = \frac{V_{SENSE,FS}}{I_{MAX}} = \frac{102.4 \text{ mV}}{I_{MAX}}. \quad (2.1)$$

To prevent saturation, a scaling factor,  $K_H$ , is applied to leave some headroom:

$$R_{SHUNT} = K_H \frac{102.4 \text{ mV}}{I_{MAX}}, \quad 0 < K_H < 1. \quad (2.2)$$

For example,  $K_H = 0.80$  reserves 20% headroom for overload, tolerance, and transient conditions.

The shunt power at maximum continuous current is

$$P_{SHUNT} = I_{MAX}^2 R_{SHUNT}. \quad (2.3)$$

The installed resistor power rating should exceed this value.

#### 2.3.1.1 Current Measurement Resolution

The LTC2945 current-sense ADC has a nominal sense-voltage resolution of 25  $\mu\text{V}$  per LSB. The current represented by one ADC count is

$$I_{LSB} = \frac{25 \mu\text{V}}{R_{SHUNT}}. \quad (2.4)$$

This relationship provides an important design tradeoff: a smaller shunt lowers power dissipation but also increases the minimum current increment represented by one ADC count.

#### 2.3.1.2 Production Shunt Design Table

Table 2.1: Production current-sense resistor calculations.

| Rail | $I_{MAX}$<br>(A) | $R_{ideal}$<br>( $\Omega$ ) | Selected $R$<br>( $\Omega$ ) | $V_{SHUNT}$<br>(mV) | $I_{LSB}$<br>(mA/LSB) | $P_{MAX}$<br>(W) |
|------|------------------|-----------------------------|------------------------------|---------------------|-----------------------|------------------|
| +5VA | 2.00             | 0.0512                      | 0.039                        | 78.0                | 0.641                 | 0.156            |
| +5VB | 2.00             | 0.0512                      | 0.039                        | 78.0                | 0.641                 | 0.156            |
| +5V  | 1.00             | 0.1024                      | 0.050                        | 50.0                | 0.500                 | 0.050            |
| -5V  | 1.00             | 0.1024                      | 0.050                        | 50.0                | 0.500                 | 0.050            |
| +15V | 1.00             | 0.1024                      | 0.050                        | 50.0                | 0.500                 | 0.050            |
| -15V | 1.00             | 0.1024                      | 0.050                        | 50.0                | 0.500                 | 0.050            |

### 2.3.1.3 Evaluation-Board Shunt Calculation

The prototype LTC2945 evaluation board was tested from a 5 V source with a load range of approximately 1 k $\Omega$  to 10 k $\Omega$ . This is a low-current bench-test condition and is separate from the production shunt design.

At the minimum load resistance, the shunt required to place the current-sense input at full scale is

$$R_{S,FS} = \frac{V_{SHUNT}R_L}{V_S - V_{SHUNT}} = \frac{(0.1024 \text{ V})(1 \text{ k}\Omega)}{5 \text{ V} - 0.1024 \text{ V}} = 20.91 \Omega. \quad (2.5)$$

Applying approximately 15% headroom gives

$$R_S = (0.85)(20.91 \Omega) \approx 17.8 \Omega. \quad (2.6)$$

At a 10 k $\Omega$  load,

$$I_{MIN} = \frac{5 \text{ V}}{10 \text{ k}\Omega + 17.8 \Omega} \approx 499.1 \mu\text{A}, \quad (2.7)$$

and therefore

$$V_{SHUNT,MIN} = I_{MIN}R_S \approx (499.1 \mu\text{A})(17.8 \Omega) \approx 8.88 \text{ mV}. \quad (2.8)$$

The prototype sense voltage therefore remains inside the LTC2945 102.4 mV full-scale range over the intended test loads.

## 2.3.2 PCB Trace and Copper-Pour Current Capacity

Power-path copper must carry the maximum rail current without excessive temperature rise or voltage drop. KiCad's track-width calculator uses a closed-form fit to IPC-2152 data. For an external trace it can be expressed as

$$\Delta T = KI^a W^b T_h^c, \quad (2.9)$$

where  $I$  is current in amperes,  $W$  is trace width in mils,  $T_h$  is copper thickness in mils, and for external tracks the KiCad coefficients are

$$K = 215.3, \quad a = 2, \quad b = -1.15, \quad c = -1.0. \quad (2.10)$$

Solving for minimum width,

$$W_{MIN} = \left( \frac{\Delta T}{KI^a T_h^c} \right)^{1/b}. \quad (2.11)$$

For 1 oz copper ( $T_h \approx 1.378$  mil) and a preliminary 10 °C allowed rise, the calculated external-trace widths are approximately 3.27 mil at 0.5 A, 10.9 mil at 1 A, and 36.4 mil at 2 A. These are thermal minimums only; the actual layout should use wider traces or copper pours where space permits.

### 2.3.3 I<sup>2</sup>C Pull-Up Resistor Sizing

The I2C bus buffer and I2C line driver both have internal active current sources that eliminate the need for external pullups. Footprints were placed for 4.7k pullups but are set as Do Not Populate.

### 2.3.4 LTC2945 Voltage Measurement Resolution

The LTC2945 rail-voltage channel has a nominal full-scale range of 102.4 V and a nominal resolution of 25 mV per LSB. The quantization-only percentage of a measured rail voltage is therefore

$$\epsilon_Q = \frac{25 \text{ mV}}{V_{RAIL}} \times 100\%. \quad (2.12)$$

For example, one count corresponds to approximately 0.42% of a 6 V rail, 0.50% of a 5 V rail, and 0.17% of a 15 V rail. This is only ADC resolution and not total measurement accuracy.

### 2.3.5 Negative-Rail Sign Convention

The -5 V and -15 V supplies are electrically isolated from the controller. The monitor can therefore be placed in the high side of each isolated rail and read a positive local differential quantity even when the system-level rail is labeled negative. Firmware applies the configured sign after acquisition:

$$V_{DISPLAY} = S_V V_{MEASURED}, \quad S_V \in \{+1, -1\}. \quad (2.13)$$

The same channel metadata is used by the local display and network interface so that a negative rail is represented consistently throughout the system.

### 2.3.6 Alarm-Limit Scaling

User alarm thresholds are stored and compared in engineering units. For a lower voltage limit  $V_L$  and upper voltage limit  $V_H$ , the normal voltage state is

$$V_L \leq V_{MEAS} \leq V_H. \quad (2.14)$$

Similarly, for lower and upper current limits,

$$I_L \leq I_{MEAS} \leq I_H. \quad (2.15)$$

If raw ADC thresholds are ever written directly to the LTC2945, the corresponding register value must be calculated from the ADC LSB size. For the current-sense channel,

$$N_I = \frac{I_{LIMIT} R_{SHUNT}}{25 \mu\text{V}/\text{LSB}}, \quad (2.16)$$

and for the rail-voltage channel,

$$N_V = \frac{V_{LIMIT}}{25 \text{ mV}/\text{LSB}}. \quad (2.17)$$

### 2.3.7 Controller-Supply Power Budget

A preliminary controller power budget was developed using datasheet maximum values where available and conservative design-current allowances for assemblies where a complete board-level maximum current is not specified. A 25% design margin is applied to the calculated load.

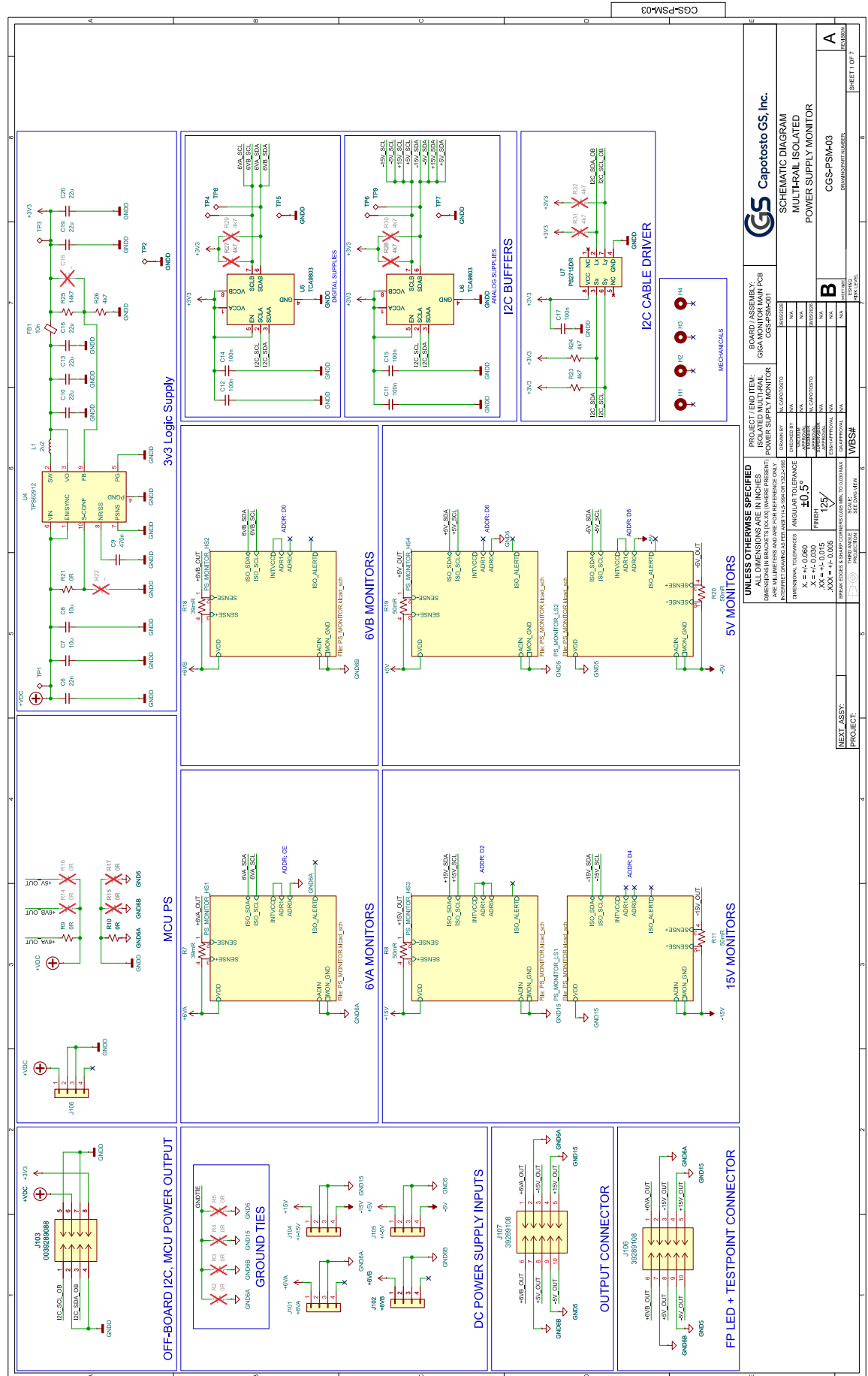
Table 2.2: Preliminary controller-side power budget.

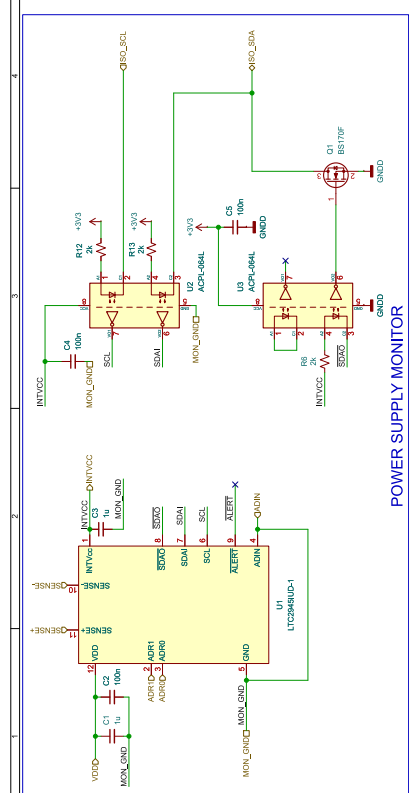
| Load                                            | Supply<br>(V) | Design Current<br>(A) | Power<br>(W) |
|-------------------------------------------------|---------------|-----------------------|--------------|
| Arduino GIGA R1 WiFi                            | 5.0           | 0.500                 | 2.500        |
| GIGA Display Shield                             | 3.3           | 0.250                 | 0.825        |
| W6100 Ethernet interface                        | 3.3           | 0.265                 | 0.875        |
| I <sup>2</sup> C isolation/controller circuitry | 3.3           | 0.065                 | 0.215        |
| Subtotal                                        | –             | –                     | 4.415        |
| 25% design margin                               | –             | –                     | 1.104        |
| <b>Total</b>                                    | –             | –                     | <b>5.519</b> |

## 2.4 Preliminary Design Schematics

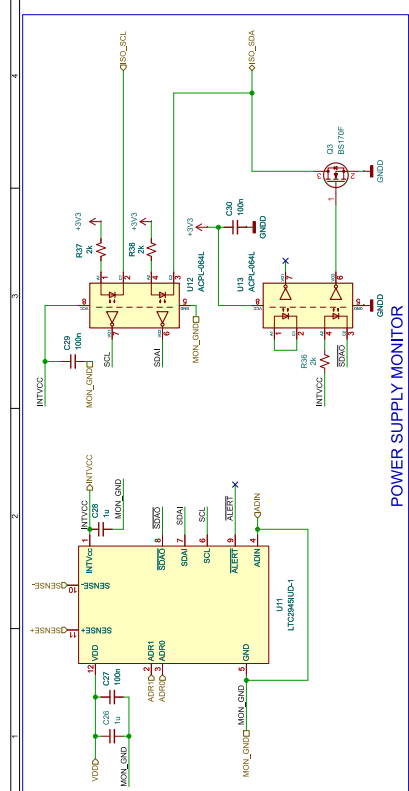
The preliminary hardware schematics are maintained in KiCad and are included below directly from the plotted PDF files in the project hardware directory.

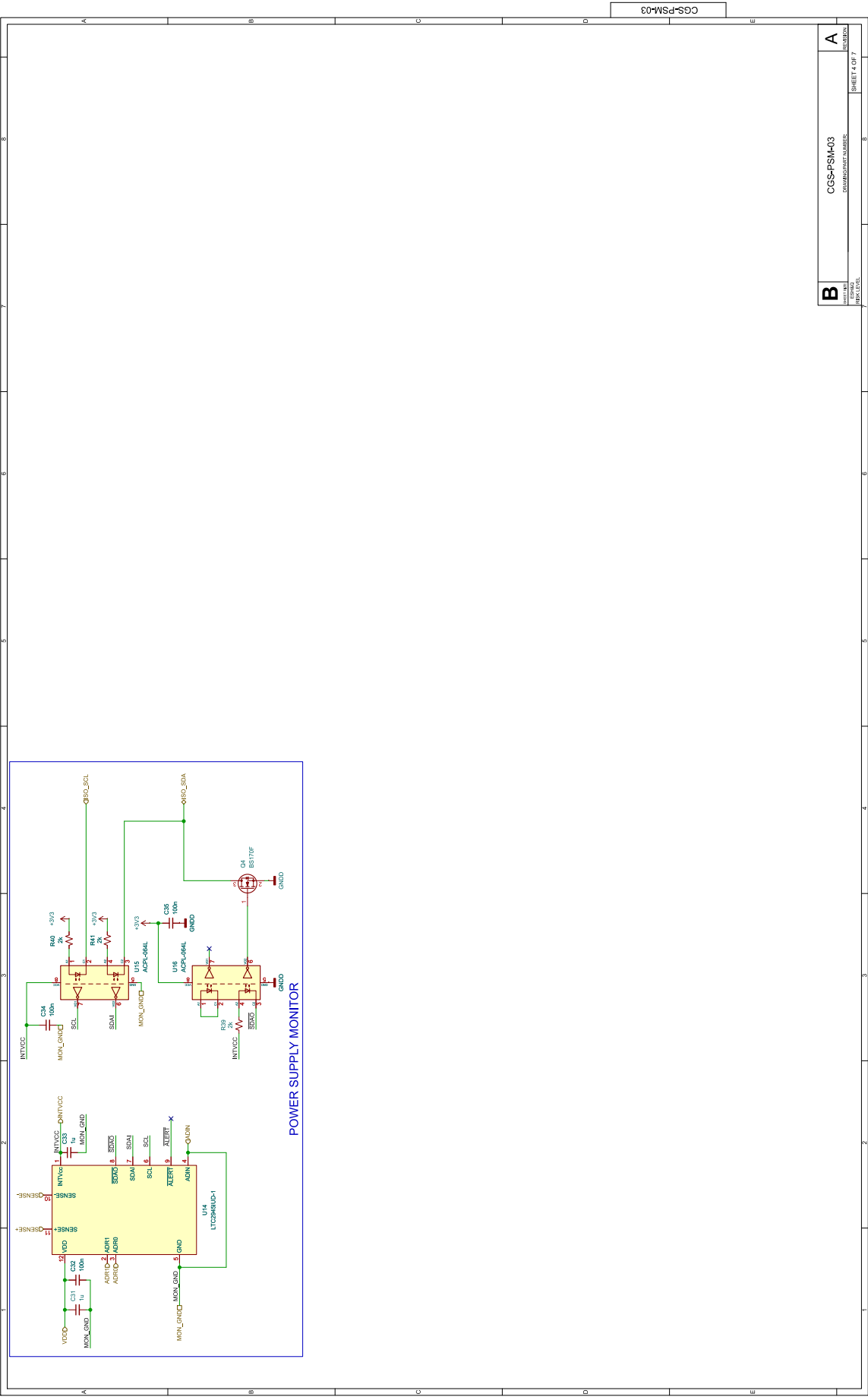
### 2.4.1 Sensor Board Schematic





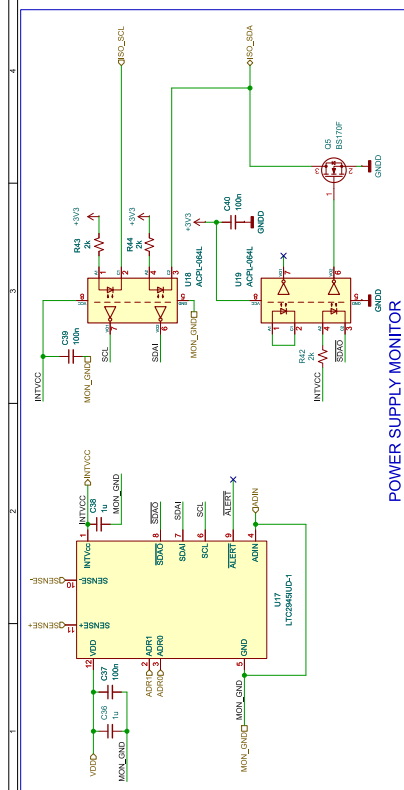




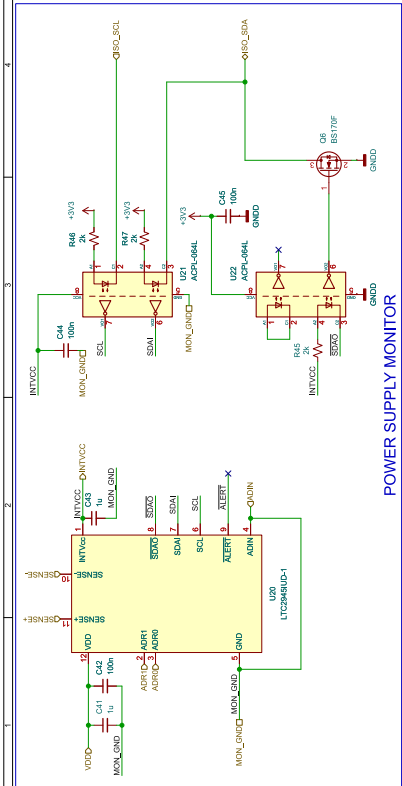


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|   |            |              |



POWER SUPPLY MONITOR



POWER SUPPLY MONITOR

CGS-PSM-03

### 2.4.2 Arduino Shield Board Schematic



## 2.5 Software Flowcharts

The software for the Multi-Rail Isolated Power Supply Monitoring System is maintained with the hardware design files in the project Git repository:

[https://github.com/capotosto/PS\\_Monitor](https://github.com/capotosto/PS_Monitor)

The repository provides version control for the Arduino GIGA firmware, hardware-design files, and EPICS IOC software used by the monitoring system. The embedded firmware is maintained under the **firmware** portion of the repository, while the EPICS IOC and associated database/configuration files are maintained under **epics-ioc**. The flowcharts in this section describe the major software operations implemented by these portions of the project.

The Arduino GIGA firmware performs system initialization, acquires voltage and current measurements from the six LTC2945 monitor channels, evaluates alarm conditions, updates the local display, and publishes data through the Ethernet interface. The primary firmware execution sequence is:

1. Initialize the GIGA Display, persistent settings, Ethernet interface, I<sup>2</sup>C interface, and channel configuration.
2. Load stored network parameters and voltage/current alarm limits from persistent storage.
3. Initialize communication with each of the six LTC2945 monitoring channels.
4. Enter the continuous acquisition and communication loop.
5. Poll each LTC2945 channel for voltage and current measurements.
6. Verify I<sup>2</sup>C communication status for each channel.
7. If communication fails, mark the associated channel with an I<sup>2</sup>C error and prevent invalid measurement data from being used.
8. Convert valid raw measurement data to engineering units and apply the appropriate channel polarity and calibration/scaling information.
9. Compare each voltage and current measurement with its configured upper and lower alarm limits.
10. Update the GIGA Display when measurement values, alarm states, or communication states change.
11. Publish the current measurements and status information through the Ethernet PSC protocol interface.
12. Service incoming configuration requests and store modified settings when required.
13. Repeat the acquisition and communication sequence continuously.

### 2.5.1 Arduino GIGA Firmware Flowchart

Figure 2.2 shows the top-level execution flow of the Arduino GIGA firmware. Following initialization, the controller continuously acquires data from each monitoring channel, verifies communication, converts valid measurements to engineering units, evaluates alarm limits, updates the local display, and services the Ethernet/PSC interface.

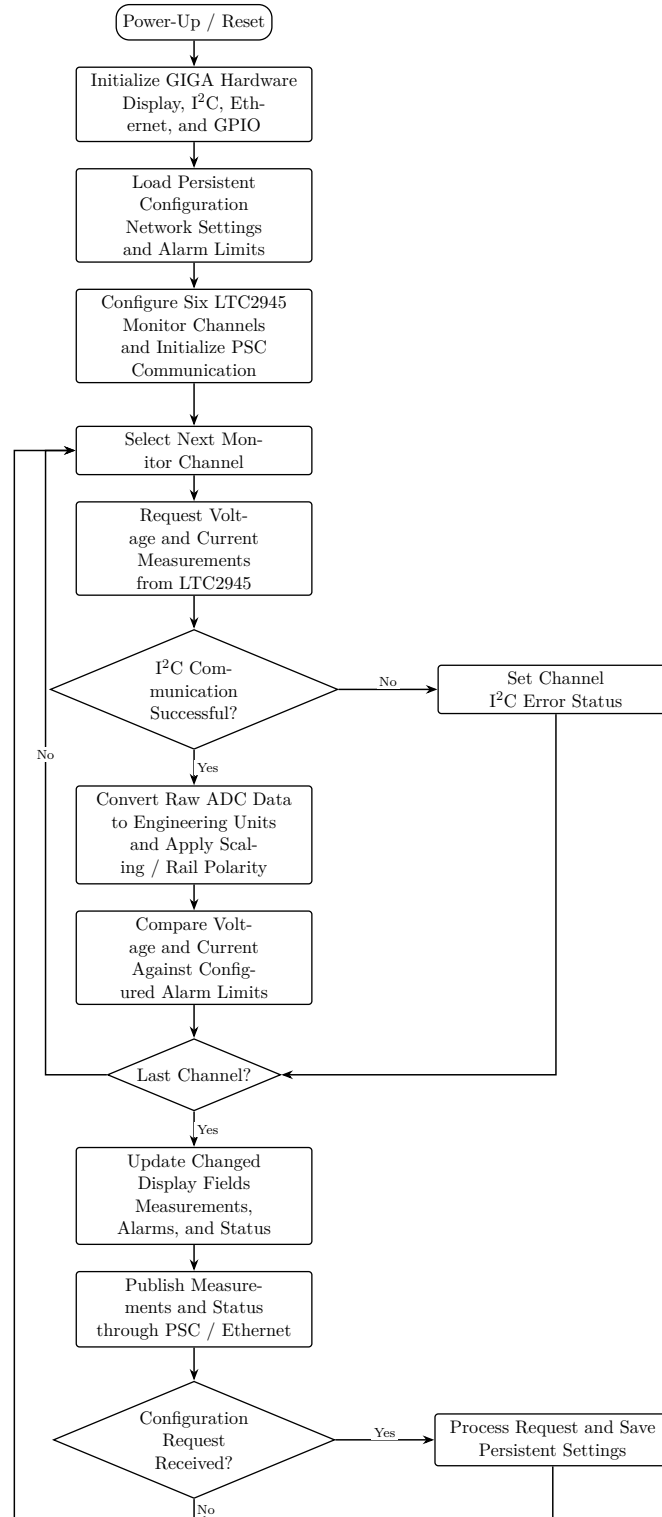


Figure 2.2: Top-level Arduino GIGA firmware flowchart. Following initialization, the controller continuously acquires measurements from the six LTC2945 monitoring channels, validates I<sup>2</sup>C communication, converts valid data to engineering units, evaluates configured alarm limits, updates the local display, publishes measurements through the PSC/Ethernet interface, and services configuration requests.

## 2.5.2 PSC and EPICS Communication Flow

The Arduino GIGA does not directly implement EPICS Channel Access. Instead, measurement and status information is transmitted through the Ethernet interface using the PSC protocol. The EPICS IOC uses PSCDriver to communicate with the GIGA, maps the received values into EPICS process variables, and makes those process variables available to the CS-Studio Phoebus operator interface.

This separation allows the embedded controller to remain responsible for hardware acquisition and local alarm evaluation while the IOC provides the interface between the embedded hardware and the facility control system. Figure 2.3 summarizes this software data path.

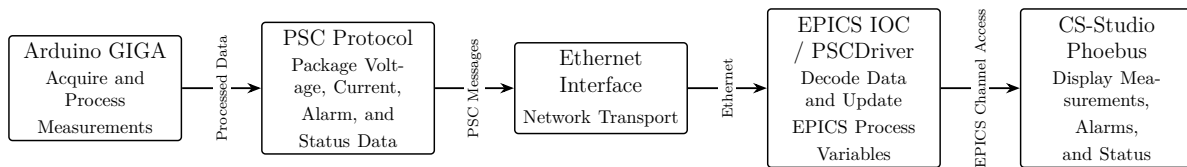


Figure 2.3: Software communication path from the Arduino GIGA firmware through PSCDriver and the EPICS IOC to the CS-Studio Phoebus operator interface.

## 2.5.3 Software Configuration Management

Firmware, IOC configuration files, database definitions, and operator-interface files are maintained under Git version control so that hardware and software revisions can be associated with a defined project baseline. The current software implementation and revision history are available in the `PS_Monitor` repository at:

[https://github.com/capotosto/PS\\_Monitor](https://github.com/capotosto/PS_Monitor)

Changes made during implementation and testing are committed to the repository, providing traceability between the preliminary design presented in this report and the final implemented system. The repository therefore serves as the controlled source for the firmware and control-system software associated with the project.

## 2.6 Preliminary Component List / Bill of Materials

The current bills of materials for the Sensor Board and Arduino Shield Board are maintained with their respective KiCad hardware-design files in the project `hardware` directory. They are included below for reference.

2.6.1 Sensor Board Bill of Materials

| BOM: Power Sensor Board             |                    |     |     | Production                 |                             | 1 Tape and Reel/Reeling Fees All: No |     | Add 10% spares for components excluding modules |           |                |  |
|-------------------------------------|--------------------|-----|-----|----------------------------|-----------------------------|--------------------------------------|-----|-------------------------------------------------|-----------|----------------|--|
| Reference                           | Value              | DNP | MPN | Manufacturer               | DPN                         | Distributor                          | Qty | Qty+10%                                         | Order Qty | Extended Price |  |
| C1,C3,C21,C23,C26,C28,C31,U100n     | CL10A105K88NNNC    |     |     | Samsong Electro-Mechanics  | 1276-1860-6-ND              | Digkey                               | 12  | 14                                              | 100       | \$3.77         |  |
| C2,C4,C5,C11,C12,C14,C15,U100n      | GRM155R61H104KE14D |     |     | Murata Electronics         | 490-10697-6-ND              | Digkey                               | 23  | 26                                              | 100       | \$1.41         |  |
| C6                                  | 22n                |     |     | Murata Electronics         | 490-1305-6-ND               | Digkey                               | 1   | 2                                               | 100       | \$1.01         |  |
| C7,C7,C8                            | 10u                |     |     | TDK                        | 445-181600-6-ND             | Digkey                               | 2   | 3                                               | 10        | \$3.58         |  |
| C9                                  | 470n               |     |     | TDK                        | 445-8020-6-ND               | Digkey                               | 1   | 2                                               | 100       | \$3.17         |  |
| C10,C13,C16,C19,C20                 | GMC21X7S226M10NT   |     |     | Cat-Chip Electronics, Inc. | 4713-GMC21X7S226M10NTDKR-ND | Digkey                               | 5   | 6                                               | 15        | \$11.30        |  |
| C18                                 | DNP                |     |     |                            |                             | Digkey                               | 1   | 2                                               |           |                |  |
| F81                                 | 10n                |     |     | Murata Electronics         | 490-BLE18PS080S1NDDKR-ND    | Digkey                               | 1   | 2                                               | 10        | \$2.30         |  |
| H1,H2,H3,H4                         | MountingHole_Pad   |     |     |                            |                             |                                      | 4   | 5                                               |           |                |  |
| J101                                | +6V/A              |     |     | Molex                      | WM14518-ND                  | Digkey                               | 1   | 2                                               | 1         | \$2.70         |  |
| J102                                | +6V/B              |     |     | Molex                      | WM14518-ND                  | Digkey                               | 1   | 2                                               | 1         | \$2.70         |  |
| J103                                | 00392890088        |     |     | Molex                      | WM3835-ND                   | Digkey                               | 1   | 2                                               | 1         | \$4.58         |  |
| J104                                | +/-15V             |     |     | Molex                      | WM14518-ND                  | Digkey                               | 1   | 2                                               | 1         | \$2.70         |  |
| J105                                | +/-5V              |     |     | Molex                      | WM14518-ND                  | Digkey                               | 1   | 2                                               | 1         | \$2.70         |  |
| J106,J107                           | 39289108           |     |     | Molex                      | 23-0039289108-ND            | Digkey                               | 2   | 3                                               | 1         | \$5.02         |  |
| J108                                | Conn_01x04         |     |     | Molex                      | WM14518-ND                  | Digkey                               | 1   | 2                                               | 1         | \$2.70         |  |
| L1                                  | 2u2                |     |     | Wurth Elektronik           | 732-8232-6-ND               | Digkey                               | 1   | 2                                               | 3         | \$5.82         |  |
| Q1,Q2,Q3,Q4,Q5,Q6                   | BS170F             |     |     | onsemi                     | NTA7002NT1GOSDKR-ND         | Digkey                               | 6   | 7                                               | 25        | \$4.43         |  |
| R2,R3,R4,R5,R14,R15,R16,R17,OR      | DNP                |     |     | 5108 Keystone Electronics  | 36-5108DKR-ND               | Digkey                               | 8   | 9                                               |           |                |  |
| R6,R12,R13,R33,R34,R35,R36,2k       | ERL-25KF2001X      |     |     | Panasonic Industry         | P2.00KLDKR-ND               | Digkey                               | 18  | 20                                              | 100       | \$1.64         |  |
| R7,R18                              | 39mR               |     |     | Ohmite                     | 273-LVK20R039FERDKR-ND      | Ohmite                               | 2   | 3                                               | 5         | \$9.45         |  |
| R8,R11,R19,R20                      | 50mR               |     |     | Ohmite                     | 273-LVK20R050FERDKR-ND      | Ohmite                               | 4   | 5                                               | 5         | \$9.45         |  |
| R9,R10                              | 0R                 |     |     | 5108 Keystone Electronics  | 36-5108DKR-ND               | Digkey                               | 2   | 3                                               | 10        | \$2.01         |  |
| R21                                 | 0R                 |     |     | Vishay Dale                | 541-0.0IDKR-ND              | Digkey                               | 1   | 2                                               | 100       | \$0.99         |  |
| R22                                 | ~                  | DNP |     |                            |                             | Digkey                               | 1   | 2                                               |           |                |  |
| R23,R24                             | 4k7                |     |     | Panasonic Industry         | 541-4.70KLCI-ND             | Digkey                               | 2   | 3                                               | 100       | \$1.57         |  |
| R25                                 | 14k7               |     |     | Panasonic Industry         | P17731CT-ND                 | Digkey                               | 1   | 2                                               | 10        | \$4.52         |  |
| R26                                 | 4k7                |     |     | Panasonic Industry         | P4.7KCPCT-ND                | Digkey                               | 1   | 2                                               | 10        | \$4.99         |  |
| R27,R28,R29,R30,R31,R32             | 4k7                | DNP |     | Vishay Dale                | 541-4.70KLCI-ND             | Digkey                               | 6   | 7                                               |           |                |  |
| TP1,TP3                             | 5000               |     |     | 5000 Keystone              | 36-5000-ND                  | Digkey                               | 2   | 3                                               | 10        | \$2.04         |  |
| TP2                                 | 5001               |     |     | 5001 Keystone              | 36-5001-ND                  | Digkey                               | 1   | 2                                               | 10        | \$2.03         |  |
| TP4,TP5,TP6,TP7,TP8,TP9             | 5015               |     |     | 5015 Keystone              | 36-5015CT-ND                | Digkey                               | 6   | 7                                               | 25        | \$8.29         |  |
| U1,U8,U11,U14,U17,U20               | LTC2945IUD#TRPBF   |     |     | Analog Devices Inc.        | LTC2945IUD#TRPBKFR-ND       | Digkey                               | 6   | 7                                               | 10        | \$110.97       |  |
| U2,U3,U9,U10,U12,U13,U15,LACPL-064L | ACPL-064L-500E     |     |     | Broadcom Limited           | 516-2614-6-ND               | Digkey                               | 12  | 14                                              | 15        | \$56.27        |  |
| U4                                  | TPS62912           |     |     | Texas Instruments          | 595-TPS62912RPUR            | Mouser                               | 1   | 2                                               | 3         | \$7.77         |  |
| U5,U6                               | TCA9803            |     |     | Texas Instruments          | 296-47829-6-ND              | Digkey                               | 2   | 3                                               | 10        | \$6.84         |  |
| U7                                  | P82715DR           |     |     | Texas Instruments          | 296-24729-6-ND              | Digkey                               | 1   | 2                                               | 5         | \$13.55        |  |
| TOTAL COST:                         |                    |     |     |                            |                             |                                      |     |                                                 |           | \$302.26       |  |

2.6.2 Arduino Shield Board Bill of Materials

| BOM: Arduino Shield Board                                                         |                       |       |                     | Production                  |                              | 1 Tape and Reel/Reeling Fees All: No |             | Add 10% spares for components excluding modules |           |                |
|-----------------------------------------------------------------------------------|-----------------------|-------|---------------------|-----------------------------|------------------------------|--------------------------------------|-------------|-------------------------------------------------|-----------|----------------|
| Reference                                                                         | Value                 | DNP   | MPN                 | Manufacturer                | DPN                          | Distributor                          | Qty         | Qty+10%                                         | Order Qty | Extended Price |
| A201                                                                              | Arduino, Giga_R1_WiFi |       | ABX00063            | Arduino                     | 1050-ABX00063-ND             | Digiteky                             | 1           | 2                                               | 1         | \$69.60        |
| C6                                                                                | 22n                   | DNP   | GRM155R71H22X4A01D  | Murata Electronics          | 490-1305-6-ND                | Digiteky                             | 1           | 2                                               | 0         |                |
| C7,C8                                                                             | 10u                   | DNP   | C2012X7S1E106K125AC | TDK                         | 445-181600-6-ND              | Digiteky                             | 2           | 3                                               | 0         |                |
| C9                                                                                | 470n                  | DNP   | C1005X5R1E474K050BB |                             | 445-8020-6-ND                | Digiteky                             | 1           | 2                                               | 0         |                |
| C10, C13, C16, C19, C20                                                           | 22u                   | DNP   | GMC21X7S226M10NT    | Cal-Chip Electronics, Inc.  | 4713-GMC21X7S226M10NTDKR-ND  | Digiteky                             | 5           | 6                                               | 0         |                |
| C17, C201, C202, C203, C211, C215, C216, C217, C220, C221, C223, C224, C225, C227 | 100n                  |       | GRM155R61H104KE14D  | Murata Electronics          | 490-10697-6-ND               | Digiteky                             | 14          | 16                                              | 100       | \$1.41         |
| C218                                                                              | 3u3                   |       |                     |                             |                              | Digiteky                             | 1           | 2                                               |           |                |
| C219                                                                              | 1n                    |       | 1206GC102KAT1A      | KYOCERA AVX                 | 478-2956-6-ND                | Digiteky                             | 1           | 2                                               | 10        | \$1.43         |
| C222, C226                                                                        | 18p                   |       | GRM1555CH180JA01D   | Murata Electronics          | 490-5858-6-ND                | Digiteky                             | 2           | 3                                               | 100       | \$1.43         |
| F81                                                                               | 10n                   | DNP   | BLE18PS080SN1D      | Murata Electronics          | 490-BLE18PS080SN1DDKR-ND     | Digiteky                             | 1           | 2                                               |           |                |
| FB201, FB202, FB203                                                               | 120R                  |       | BBUP00160805121Y00  | Pulse Electronics           | 553-BBUP00160805121Y00CT-ND  | Digiteky                             | 3           | 4                                               | 25        | \$1.97         |
| J201                                                                              | SI-46014-F            |       | SI-46014-F          | Bel Fuse Inc.               | 5923-SI-46014-F-ND           | Digiteky                             | 1           | 2                                               | 1         | \$5.81         |
| J309                                                                              | 0039289088            |       | 0039289088          | Molex                       | WM3835-ND                    | Digiteky                             | 1           | 2                                               | 1         | \$4.58         |
| L1                                                                                | 2u2                   | DNP   | 74438356022         | Wurth Elektronik            | 732-8232-6-ND                | Digiteky                             | 1           | 2                                               | 3         | \$5.82         |
| N101                                                                              | Giga_Display          |       | ASX00039            | Arduino                     | 1050-ASX00039-ND             | Digiteky                             | 1           | 2                                               | 1         | \$63.60        |
| N201                                                                              | Connector_Set         |       | 85                  | Adafruit Industries         | 1528-1074-ND                 | Digiteky                             | 1           | 2                                               | 2         | \$3.90         |
| R21                                                                               | 0R                    | DNP   | ERJ-2RKF4701X       | Panasonic Industry          | P4.70KLDKR-ND                | Digiteky                             | 1           | 2                                               |           |                |
| R23, R24, R205, R206, R207                                                        | 4k7                   |       | CRCW04024K70FKED    | Vishay Dale                 | 541-4-70KLCF-ND              | Digiteky                             | 5           | 6                                               | 100       | \$1.57         |
| R25                                                                               | 15k2                  | DNP   | ERJ-2RKF4701X       | Panasonic Industry          | P4.70KLDKR-ND                | Digiteky                             | 1           | 2                                               |           |                |
| R26                                                                               | 4k87                  | DNP   | ERJ-2RKF4701X       | Panasonic Industry          | P4.70KLDKR-ND                | Digiteky                             | 1           | 2                                               |           |                |
| R31, R32                                                                          | 4k7                   | DNP   | ERJ-2RKF4701X       | Panasonic Industry          | P4.70KLDKR-ND                | Digiteky                             | 2           | 3                                               |           |                |
| R101, R102, R103, R104, R105                                                      | 10k                   |       |                     |                             |                              | Digiteky                             | 5           | 6                                               |           |                |
| R208                                                                              | 12k3                  |       | RN73R1ETTP1232B25   | KOA Speer Electronics, Inc. | 2019-RN73R1ETTP1232B25DKR-ND | Digiteky                             | 1           | 2                                               | 50        | \$3.02         |
| R209, R210, R215, R216                                                            | 49R9                  |       | CRCW040249R9FKED    | Vishay Dale                 | 541-49.9LDKR-ND              | Digiteky                             | 4           | 5                                               | 100       | \$1.57         |
| R211, R212, R213, R214                                                            | 3R3                   |       | CRCW04023R30FKED    | Vishay Dale                 | 541-3-30LLDKR-ND             | Digiteky                             | 4           | 5                                               | 100       | \$2.03         |
| R217                                                                              | 1M                    |       | CRCW04021M00FKED    | Vishay Dale                 | 541-1.00MLDKR-ND             | Digiteky                             | 1           | 2                                               | 100       | \$1.57         |
| R218, R219                                                                        | 470R                  |       | ERJ-2RKF4700X       | Panasonic Industry          | P470LDKR-ND                  | Digiteky                             | 2           | 3                                               | 100       | \$1.64         |
| R220                                                                              | 0R                    |       | CRCW04020000Z0ED    | Vishay Dale                 | 541-0.0IDKR-ND               | Digiteky                             | 1           | 2                                               | 100       | \$0.99         |
| TP1, TP3                                                                          |                       | 2 DNP | 5000                | Keystone                    | 36-5000-ND                   | Digiteky                             | 2           | 3                                               |           |                |
| TP2                                                                               |                       | 1 DNP | 5001                | Keystone                    | 36-5001-ND                   | Digiteky                             | 1           | 2                                               |           |                |
| TP101                                                                             |                       | 1     | 5000                | Keystone                    | 36-5000-ND                   | Digiteky                             | 1           | 2                                               | 10        | \$2.04         |
| TP102                                                                             |                       | 1     | 5001                | Keystone                    | 36-5001-ND                   | Digiteky                             | 1           | 2                                               | 10        | \$2.03         |
| TP201, TP202, TP203                                                               |                       | 3     | 5015                | Keystone                    | 36-5015CT-ND                 | Digiteky                             | 3           | 4                                               | 10        | \$3.70         |
| U4                                                                                | TPSG2912              | DNP   |                     |                             |                              |                                      | 1           | 2                                               |           |                |
| U7                                                                                | P82715DR              |       | P82B715DR           | WIZnet                      | 296-24729-6-ND               |                                      | 1           | 2                                               | 2         | \$5.42         |
| U202                                                                              | W6100-Q               |       | W6100Q              | WIZnet                      | 950-W6100Q                   | WIZnet                               | 1           | 2                                               | 2         | \$5.04         |
| Y201                                                                              | 25MHz                 |       | FL2500169           | Diodes Incorporated         | 31-FL2500169DKR-ND           | Digiteky                             | 1           | 2                                               | 2         | \$1.38         |
|                                                                                   |                       |       |                     |                             |                              |                                      | TOTAL COST: |                                                 |           | \$191.55       |

# Chapter 3

## Installing EPICS Base and IOCs

### 3.1 Introduction

A simple IOC was written to interface the Arduino Giga which monitors the six supplies. The following section details how to install EPICS Base and required modules, and compile the IOC. It is written assuming a Linux environment; this particular build was done on Ubuntu 22.04LTS.

### 3.2 EPICS Installation Overview

EPICS installations performed by M. Capotosto loosely follow the below directory structure:

- `EPICS_BASE=/epics/base` or `/epics/base_[version]` or similar
- `SUPPORT=/epics/support` – or sometimes `MODULES` instead of `SUPPORT` is used.

Modules required for the PSU monitor are:

- `pscdrv`: Portable Streaming Controller Driver
- `epics/iocs` - each IOC is cloned into this repository
- `epics/systemd-softioc` - This is a utility for managing IOCs as a system service

### 3.3 Installing EPICS Base and Required Modules

1. Install Dependencies for Build      Install git, build-essential, and perl:  

```
sudo apt -y install git build-essential perl libtirpc-dev re2c libevent-dev
autoconf automake libtool pkg-config
```

2. Create the EPICS user group, and the empty EPICS directory:

```
sudo groupadd epics
sudo usermod -aG epics $USER
newgrp epics
sudo mkdir -p /epics
sudo chown $USER:epics /epics
sudo chmod 2775 /epics
```

3. EPICS Base is cloned into /epics/base via:

```
git clone -recursive -b 7.0
https://github.com/epics-base/epics-base.git /epics/base
cd /epics/base
```

4. Build EPICS:

```
make clean
make
```

5. Create the remaining directory structure:

```
mkdir -p /epics/iocs /epics/modules
```

6. Clone PSC Driver:

- (a) `cd /epics/modules`
- (b) `git clone https://github.com/mdavidsaver/pscdrv.git pscdrv`
- (c) `cd /epics/modules/pscdrv`
- (d) Set up required files in `configure/RELEASE`, `configure/RELEASE.LOCAL`
- (e) Build with `make clean`, and `make`

7. Clone Manage IOCs Utility:

- (a) `cd /epics`
- (b) `git clone https://github.com/NSLS2/systemd-softioc.git`
- (c) Navigate to the repo directory `cd systemd-softioc`
- (d) Make the install script executable and run:
 

```
sudo chmod +x install.sh
sudo ./install.sh
sudo usermod -aG epics softioc
newgrp
sudo chown softioc:epics /epics/iocs
sudo chmod 2775 /epics/iocs
```

### 3.4 Installing the PSU Monitor IOC

1. Change to IOC directory `cd /epics/iocs`
2. Clone the repo:  
`git clone https://github.com/capotosto/PS_Monitor psuMonitorIOC`
3. `cd psuMonitorIOC`
4. Create the `RELEASE.local` file, `touch configure/RELEASE.local`
5. `echo -e "EPICS_BASE = /epics/base\nPSCDRV = /epics/modules/pscdrv\n"`

6. Build the IOC: `make clean, make`
7. Update the `st.cmd` file as needed: `cd iocBoot/iocpsuMonitor`
8. Create the `st.cmd` file used by `systemd-softioc`:

```
echo -e '#!/bin/bash\n\n
cd "/epics/iocs/psuMonitorIOC/iocBoot/iocpsuMonitor"\n./st.cmd' > st.cmd
```

9. Make the new `st.cmd` executable: `chmod +x st.cmd`
10. Find the next available port: `manage-iocs nextport`
11. Configure this in the `config` file: `nano config`
12. Set the `NAME=`, `PORT=` your next port, `USER=softioc`, and `HOST=$HOSTNAME`
13. Update the directory path in the root-level `st.cmd` to the `iocBoot/iocpsuMonitor/st.cmd`
14. Enroll in `manage-iocs` as a service: `sudo manage-iocs install psuMonitorIOC`
15. Enable auto-start on boot: `sudo manage-iocs enable psuMonitorIOC`
16. Check the status, it should be running! `manage-iocs status`
17. And if you want other information on the ioc: `manage-iocs report`



# Chapter 4

## CS-Studio - Phoebus

### 4.1 Introduction

CS-Studio Phoebus is installed using precompiled binaries. This application requires a Java runtime environment be installed.

The 'BOB' files - the user interface panels, are located in the IOC's directories, `/epics/iocs/PSCtest/gui` and `/epics/iocs/zpsc_ioc/gui`

Behavior of some features of Phoebus must be modified by the shell script used to launch Phoebus, an example is included with the below installation instructions to modify the behavior of the 'Home' button, used to bring up the custom 'PSC Launcher' panel using the Home button, instead of having to manually navigate to the file when needed.

### 4.2 Installing Phoebus

1. If not installed, install Java, e.g.:  
`sudo apt update && sudo apt install openjdk-17-jdk`
2. Create the directory: `sudo mkdir /opt/phoebus`  
`sudo chown $USER:epics /opt/phoebus`  
`chmod 2775 /opt/phoebus`
3. Download Phoebus and extract to the `/opt/phoebus` directory  
<https://github.com/ControlSystemStudio/phoebus/releases/download/v4.7.4/phoebus-4.7.4-linux.tar.gz>

```
cd /opt/phoebus
```

```
wget https://github.com/ControlSystemStudio/phoebus/releases/download/  
v4.7.4/phoebus-4.7.4-linux.tar.gz
```

```
tar -xvzf phoebus-4.7.4-linux.tar.gz -C . -strip-components=1
```

4. Make the shell script executable: `chmod +x phoebus.sh`
5. Alias Phoebus to run via command `run-phoebus`:

```
echo 'alias run-phoebus="/opt/phoebus/phoebus.sh"' » ~/.bashrc  
source ~/.bashrc
```
6. Now test! Try `run-phoebus` from a terminal.

# Chapter 5

## Prototyping and Development

### 5.1 Introduction

### 5.2 Testing the LTC2945

The LTC2945 testing was carried out using a Linear Tech eval board, DEMO CIRCUIT 1697A. Connections and modifications were made as follows:

- VIN: tied to Arduino +5V
- GND: tied to Arduino GND
- ADIN: Tied to GND
- SCL: Tied to Arduino Pin D8 I2C2 SCL
- SDAI/SDAO (JP2 set to TIE): Arduino Pin D9 I2C2 SDA
- VPU: Tied to Arduino +3v3
- VOUT: Float or tied to dummy load ( $1\text{k}\Omega$  to  $10\text{k}\Omega$ )
- $20\text{m}\Omega$  shunt replaced with  $17\Omega$  shunt for testing with  $\sim 5\text{k}\Omega$  load
- Installed  $4.7\text{k}\Omega$  pull up on I2C lines via R4, R5 footprints on EVB

### 5.2.1 Calculating Shunt Value for Test

Using the 5V from the arduino as a source, and a load ranging 1k to 10k ohms, we calculate:

$$R_{L,min} = 1k\Omega$$

$$V_{shunt} = V_S \frac{R_S}{R_L + R_S}$$

$$R_S = \frac{V_{shunt} R_L}{V_s - V_{shunt}}$$

$$V_{shunt} = 0.1024V$$

$$R_{S,fullscale} = \frac{(0.1024V)(1k\Omega)}{5V - 0.1024V} = 20.91\Omega$$

Adding some headroom, say, 15%, and going to an E96 resistor value...

$$R_S = 0.85(20.91) = 17.8\Omega$$

Verifying for the high end of the range:

$$I_{min} = \frac{5V}{10k\Omega + 17.8\Omega} = 499.1\mu A$$

$$V_{shunt,min} = (499.1\mu A)(17.8\Omega) = 8.88mV$$

The shunt will range from 8.88mV to 87.44mV, within the ADC's usable range of 0V to 102.4mV.

### 5.2.2 Pictures of Evaluation Board, Display, and Phoebus Panels

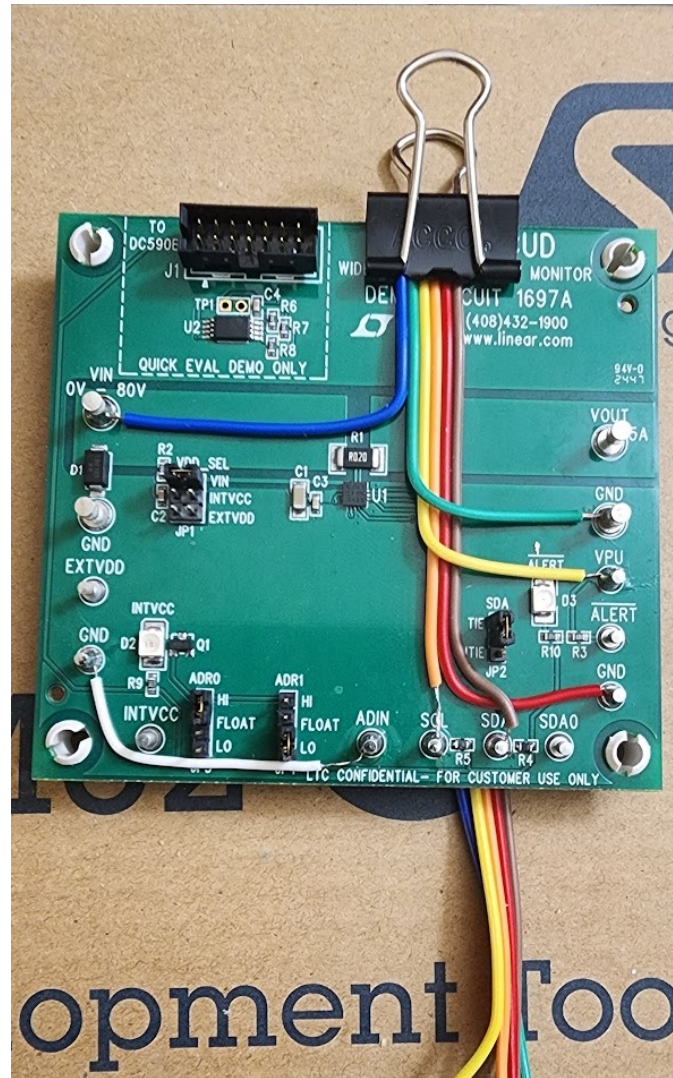


Figure 5.1: DEMO CIRCUIT 1697A Eval Board with Original Shunt, fly leads to connect to Arduino.



Figure 5.2: DEMO CIRCUIT 1697A Eval Board with  $17\Omega$  Shunt, fly leads to connect to Arduino, and  $4.7k\Omega$  test load.

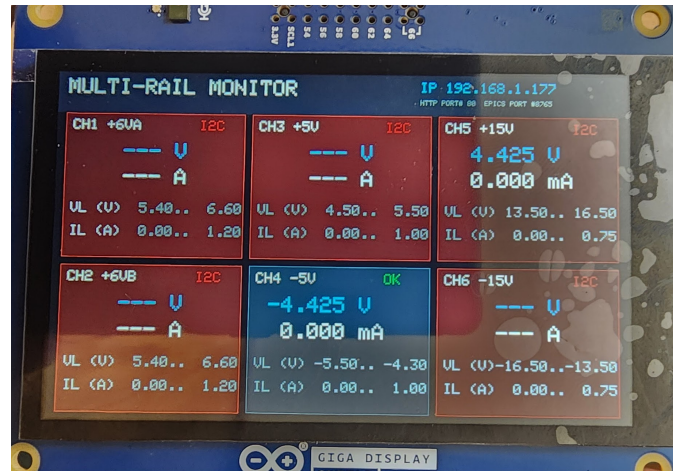


Figure 5.3: Arduino Giga with GIGA DISPLAY SHIELD showing dummy -5V rail being monitored, without alarm

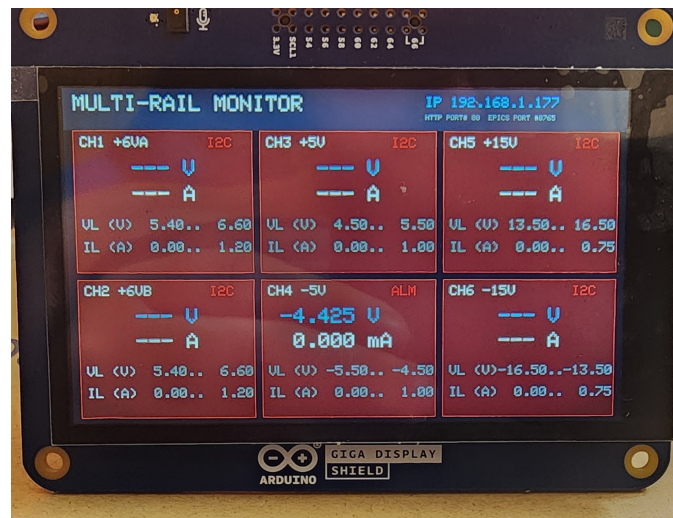


Figure 5.4: Arduino Giga with GIGA DISPLAY SHIELD showing dummy -5V rail being monitored, with UVL alarm, other channels I2C ALARM.

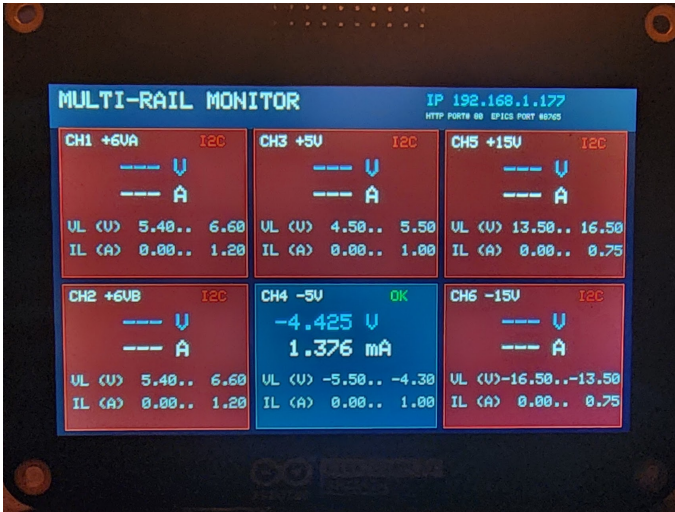


Figure 5.5: Arduino Giga with GIGA DISPLAY SHIELD showing dummy -5V rail being monitored, without alarm, 3.3kΩ Load



Figure 5.6: Arduino Giga with GIGA DISPLAY SHIELD showing dummy -5V rail being monitored, without alarm, 3.3kΩ Load



Figure 5.7: Arduino Giga with GIGA DISPLAY SHIELD connected to an LTC2945 eval board

